

To whom it may concern,

The pharmaceutical industry should not have the right to possess living cells in the name of profit, especially when millions of disease and cancer patients rely on these new treatments. Stem cell treatments, along with CRISPR techniques that target and replace specific RNA sequences, has the vast potential to drastically mitigate and even cure a diverse array of such conditions. Extraction of one's own body cells should not be classified as a drug, and will slow the progress of science if these guidelines go through because of complex patents and licensing.

Even though stem cells and their therapies are at its infancy, research into this field has resulted in drastic changes. For instance, in the published research paper from Stanford University's school of Medicine (Jackson, Kathyjo Ann, and And Tiejuan Mi.), a team of researchers discovered its significant implications after injecting pluripotent stem cells extracted from bone marrow in 6 irradiated mice. The results were remarkable, in that all of the recipients had high levels of recovery. Reintroducing these modified stem cells back to its recipients was not a major hassle, as the stem cell's DNA code closely resembles its host. This shows that these stem cells still belong to the host recipient (in that the stem cells were not rejected by its host). Licensing specific DNA codes and other vital treatments to pharmaceutical companies would introduce more issues than what is foreseen, and has already caused backlash from the general public because of such licensing (refer to Martin Shkreli's Daraprim and Mylan's Epipen incidents) and their resulting dominant monopolies (Daraprim pill from \$13.50 to \$750 & Mylan Epipen from \$94 to \$600, refer to links below).

Overall, this policy needs to be revisioned where such policies should not belong to the select few. Stem cells need to be treated as the property of the patient, especially when extracting them from the body. A better solution would be to revise the policies that favor towards the ~90% of Americans that otherwise could not afford these breakthroughs in stem cell research.

Signed,
Fernando Lorenzo
25th of September, 2016

Jackson, Kathyjo Ann, and And Tiejuan Mi. "Kathyjo Ann Jackson." Hematopoietic Potential of Stem Cells Isolated from Murine Skeletal Muscle, Regulations.gov, 1 Sept. 1999,
<http://www.pnas.org/content/96/25/14482.short>.

Martin Shkreli Daraprim - <http://www.towleroad.com/2015/12/shkreli-daraprim/>

Mylan CEO's Epipen testifying against US Congress -

<http://www.cnn.com/videos/cnnmoney/2016/09/23/mylan-ceo-epipen-hearing-cnnmoney.cnn>

Mylan Price Hike (Google "epipen price hike")-

<http://money.cnn.com/2016/09/24/technology/diy-epipen-affordable-alternatives/>

Daraprim Price Hike (Google "daraprim price hike")-

<http://www.marketwatch.com/story/heres-why-daraprim-still-costs-750-a-pill-2016-02-03>